Discharging H-diag V: reducing the full off-diagonal operator-norm

TECT verification-first repository · claim B2-PROPA-HLAYER

first issued 2026-06-10 · this version issued 2026-06-10 · v1.1

1. Purpose and scope

The full-operator-norm formulation note constructed the off-diagonal Hessian $\operatorname{Hess} F|_{G_*} = E + B_{\mathrm{od}}$ and isolated the obstruction to the worst-direction certificate $\|E^{-1/2} B_{\mathrm{od}} E^{-1/2}\| < 1$ as the sign of the exchange (Fock) block, naming it a SPECTRAL residual (the exchange-Hessian eigenvalues). This note sharpens the residual in three honest steps – a diagonality lemma removing the condensate-direction caveat, a triangle inequality isolating the exchange block with an explicit budget, and a row-sum bound replacing the eigenvalue spectrum by a single competitor-dependent scalar threshold. None is a closure; the scalar that remains is RES-5. Scope: the operating endpoint anchor $\mu^2 = 0.005$; the full admissible crystallographic competitor class (BCC quoted as the anchor specialisation).

2. Block decomposition and the diagonality lemma

With $E = \frac{1}{2} G_*^{-1} \otimes G_*^{-1} \succ 0$ (positive-definite, diagonal in Q) the interaction Hessian splits as $B_{\mathrm{od}} = B^{\mathrm{cond}} + B^{\mathrm{exch}}$, the Hartree part carrying no off-diagonal block. From Math428 the condensate/bubble off-diagonal free-energy share is

$$\Delta F_{\mathrm{od}}(A) = -\tfrac{1}{4} \sum_{Q \neq 0} W(Q)^2 \, B(|Q|), \qquad W(Q) = 3u_{\mathrm{eff}} A^2 p_2(Q) + 5vA^4 p_4(Q),$$

a sum of INDEPENDENT channel contributions.

Diagonality lemma. *If a free-energy share is a sum of independent per-channel terms $\sum_Q f_Q(G_{QQ})$, its second variation $\delta^2 = \sum_Q f_Q''(G_{QQ})\,(\delta G_{QQ})^2$ has no $Q \neq Q'$ cross term; hence the induced δG-Hessian B^{cond} is block-diagonal in the channel index Q.* Consequently $\|E^{-1/2} B^{\mathrm{cond}} E^{-1/2}\|_{\mathrm{op}} = \max_Q$ (single-channel ratio), a worst-direction (all-directions) statement, NOT merely the condensate-direction one. The exchange block B^{exch} is genuinely off-diagonal; its sign is not fixed by $u_{\mathrm{eff}} > 0$ (bare $u = -0.86 < 0$, $u_{\mathrm{eff}} = +2.685 > 0$ via the sextic dressing).

3. The condensate block over the FULL admissible class

The constant certificate bounds the single-channel ratio by $R_{\mathrm{lead}}(Q)$, and the floor identity pins it to the additive-energy floor $K_{\mathrm{floor}} = E_+/N^2 - 1$:

$$R_{\mathrm{lead}}(Q) = \mathrm{const}\,(1 + K_{\mathrm{floor}}(Q))\, I, \qquad \mathrm{const} = \frac{9}{4}\frac{u_{\mathrm{eff}}^2 B_{\max} N}{c_{\mathrm{diag}}} = 23.2, \quad I = 2 \times 10^{-3}.$$

Operator-review correction. v1.0 quoted $R_{\mathrm{lead}} = 0.174$, which is the BCC ANCHOR value ($K_{\mathrm{floor}} = 2.75$). Over the full admissible class the floor is larger (uniform worst $K_{\mathrm{floor}} = 12.13$, T-014; non-uniform $T' \leq 13$, T-015):

$$\boxed{R_{\mathrm{lead}}^{\mathrm{BCC}} = 0.174\,(\times 5.7), \quad R_{\mathrm{lead}}^{\mathrm{unif}} = 0.609\,(\times 1.64), \quad R_{\mathrm{lead}}^{\mathrm{nonunif}} \leq 0.650\,(\times 1.54), \quad \text{all} < 1}$$

so the condensate block is worst-direction certified < 1 over the WHOLE class, but with margin $\times 1.54$, not $\times 5.7$.

4. Triangle inequality and the competitor-dependent threshold

By the triangle inequality, $\|E^{-1/2} B_{\mathrm{od}} E^{-1/2}\|_{\mathrm{op}} \le R_{\mathrm{lead}}(Q) + \rho_{\mathrm{exch}}$ with $\rho_{\mathrm{exch}} = \|E^{-1/2} B^{\mathrm{exch}} E^{-1/2}\|_{\mathrm{op}}$, so the full norm is < 1 provided $\rho_{\mathrm{exch}} < 1 - R_{\mathrm{lead}}(Q)$ (budget 0.350 at the binding non-uniform point, 0.826 at the BCC anchor). For a symmetric operator $\|M\|_{\mathrm{op}} \le \max_Q \sum_{Q'} |M_{QQ'}|$; factoring the exchange row sum as $N_{\mathrm{exch}}(Q)\, b_{\mathrm{exch}}$ ($N_{\mathrm{exch}} = \max_Q \#\{Q' : Q - Q' \in \{k_i - k_j\}\}$, combinatorial; $N_{\mathrm{exch}} = 17 \le n_{\mathrm{chans}} = 54$ for BCC) gives the **competitor-dependent closure criterion**

$$\boxed{b_{\mathrm{exch}}(Q) \;<\; b_*(Q) := \frac{1 - R_{\mathrm{lead}}(Q)}{N_{\mathrm{exch}}(Q)} \quad \forall\, Q \in \mathcal{A}_{\mathrm{adm}}}$$

with the explicit values $b_*^{\mathrm{BCC}} = 0.0486$, $b_*^{\mathrm{unif}} = 0.0230$, $b_*^{\mathrm{nonunif}} = 0.0206$ (the binding full-class target). The v1.0 single number 0.0486 was the BCC anchor only; the class target is $\sim 2.4\times$ tighter. This replaces the exchange-Hessian eigenvalue spectrum by one scalar bound per competitor.

5. What this is / is NOT

A REDUCTION (T3 formulation), not a closure: b_{exch} (the dressed Fock-bubble magnitude/sign) is the RES-5 scalar and is NOT bounded here. The increment over the formulation note is that (a) the condensate block is now worst-direction certified over the full class (diagonality lemma), (b) the residual is the exchange block ALONE with explicit competitor-dependent budget, (c) the eigenvalue spectrum is replaced by a single scalar threshold per competitor. No tier flip: B1/B2 remain T6 on {H-LAYER}.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"v1.0 used the BCC anchor R_lead and over-stated the budget."* UPHELD and corrected (operator review): the full-class margin is $\times 1.54$ ($R_{\mathrm{lead}}^{\mathrm{nonunif}} = 0.650$), the budget 0.350, and the binding threshold $b_*^{\mathrm{nonunif}} = 0.0206$. The competitor-dependent form is now the canonical statement.

(β) *"The diagonality of B^{cond} is asserted."* ADDRESSED: §2 states the diagonality lemma explicitly – a sum of independent per-channel free-energy terms has a block-diagonal δG-Hessian (no $Q \neq Q'$ cross term). The genuinely off-diagonal coupling resides entirely in the exchange block, the residual carried here.

(γ) *"N_{exch} may vary across competitors."* VALID and incorporated: the threshold $b_*(Q) = (1 - R_{\mathrm{lead}}(Q))/N_{\mathrm{exch}}(Q)$ is stated $\forall Q \in \mathcal{A}_{\mathrm{adm}}$; $N_{\mathrm{exch}} = 17$ is the BCC value, a combinatorial count to be evaluated per competitor (bounded by n_{chans}).

Quantitative sanity check (mandatory): *const* – $(9/4)u_{\mathrm{eff}}^2 B_{\max} N / c_{\mathrm{diag}} = 23.2$ reproduces the operator value; *monotonicity* – R_{lead} increases with K_{floor}: $0.174 < 0.609 < 0.650 < 1$; *threshold* – tightens correspondingly $0.0486 \to 0.0206$; *sign* – bare $u = -0.86 < 0$, dressed $u_{\mathrm{eff}} = +2.685 > 0$, so b_{exch}'s sign is genuinely unfixed (RES-5).

7. Result footer

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Result ID:                    B2-PROPA-HLAYER / H-diag discharge V (full
                             off-diagonal
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operator-norm residual reduced to the exchange
block; v1.1
full-admissible-class re-issue)

Precise statement: Hess F|G* = E + B^cond + B^exch. (i)
DIAGONALITY LEMMA:
Delta F_od = sum_Q f_Q(G_QQ) [independent
channels] =>
B^cond block-diagonal in Q => ||B^cond||_op =
max-channel
ratio <= R_lead(Q). Class values
R_lead^BCC=0.174,
R_lead^unif=0.609, R_lead^nonunif<=0.650, all<1
(margins
x5.7/x1.64/x1.54). (ii) Triangle: ||E^-1/2
B_od E^-1/2||
<=
R_lead(Q) + rho_exch, so full norm <1 provided
rho_exch <
1-R_lead(Q) (budget 0.350 binding). (iii)
Symmetric row-sum:
rho_exch <= N_exch(Q) b_exch;
COMPETITOR-DEPENDENT threshold
b_exch(Q) < b_star(Q) =
(1-R_lead(Q))/N_exch(Q), values
b_star =
0.0486(BCC)/0.0230(unif)/0.0206(nonunif
binding).

Scope: Operating endpoint mu^2=0.005; full admissible
crystallographic
class (BCC = anchor specialisation). b_exch NOT
bounded.

Dependencies: hdiag-full-operator-norm-formulation (block
decomposition);
hdiag-offdiag-constant-certificate (const,
B_max, c_diag);
res-endpoint-admissible-pin (T-014
K_floor=12.13 / T-015 T'=13);
hdiag-offdiag-additive-energy (sum p_2^2=E_+);
Math428 (W,B).

Evidence grade: ANALYTIC (reduction + diagonality lemma) +
EXECUTED
(res5_018_offdiag_gershgorin_threshold.py v1.2
5/5).

Reproduction command: python
codes/vacuum/res5_018_offdiag_gershgorin_threshold.py

Expected output: R_lead 0.174/0.609/0.650; b_star
0.0486/0.0230/0.0206;
N_exch=17; 5/5 PASS.

Falsification gate: A proof that B^cond is NOT diagonal in Q
(breaks the lemma);
or an exchange computation giving b_exch >=
b_star^nonunif=
0.0206 at operating intensity for some
admissible Q (refutes
closure via this route, leaving RES-5 the only
path).

Tier before / after: No tier flip. B1 T6, B2 T6 on {H-LAYER}. RES-1
off-diagonal
residual reduced from the exchange-Hessian
eigenvalue spectrum
to the competitor-dependent scalar b_exch(Q) <

(1-R_lead(Q))/
N_exch(Q), binding full-class target 0.0206.

No-overclaim statement: Does NOT close the full off-diagonal operator norm and does NOT flip B1/B2. The exchange scalar b_exch is RES-5. The full-class margin is x1.54 (not the BCC x5.7); the binding threshold is 0.0206 (not 0.0486). B1/B2 stay T6 on {H-LAYER}.

Next required action: Bound the dressed exchange scalar b_exch(Q) at operating intensity against b_star(Q) = (1-R_lead(Q))/N_exch(Q) for the admissible class (binding target 0.0206) = RES-5 / GAP-2.